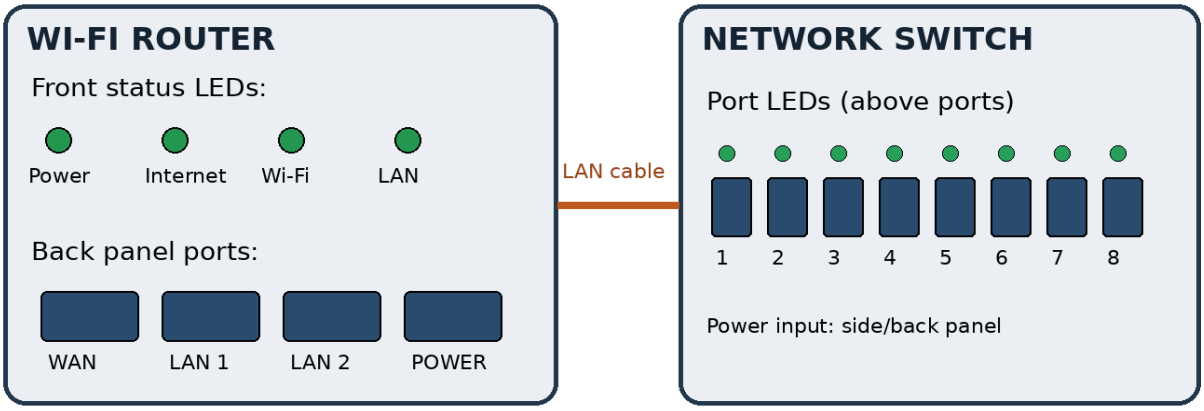
UNDERSTANDING AND MAINTENANCE OF NETWORKS

Topic: Routers and Switches

Question 1. Draw a labelled diagram of a home Wi-Fi router and a network switch.

The diagram below shows the common ports and LED indicators found on these devices. The exact position may be different on different models.

Typical Home Wi-Fi Network: Router and Switch



Router: WAN port connects to the modem/Internet Service Provider (ISP). LAN ports connect computers or a switch. The power port supplies electricity. Status LEDs show power, Internet, Wi-Fi and LAN activity.

Switch: Ethernet ports connect computers and other network devices. Small LEDs near the ports show link and data activity. The power input connects to the power adapter.

Note: A Console port is usually found on managed business switches, not on every home router or basic switch.

Question 2. List 3 key differences between routers and switches.

Point	Router	Switch
Main function	Connects different networks, such as a home network to the Internet.	Connects devices inside the same Local Area Network (LAN).
Address used	Uses IP addresses to decide where to send packets between networks.	Uses MAC addresses to send Ethernet frames to the correct device/port.
Traffic handling	Chooses a route between networks and can provide NAT, DHCP and firewall features.	Learns MAC addresses and forwards data only to the required port when possible.

Question 3. Identify the power port, at least 2 Ethernet ports and LED indicators on each device.

A. Wi-Fi Router — real product photo reference:

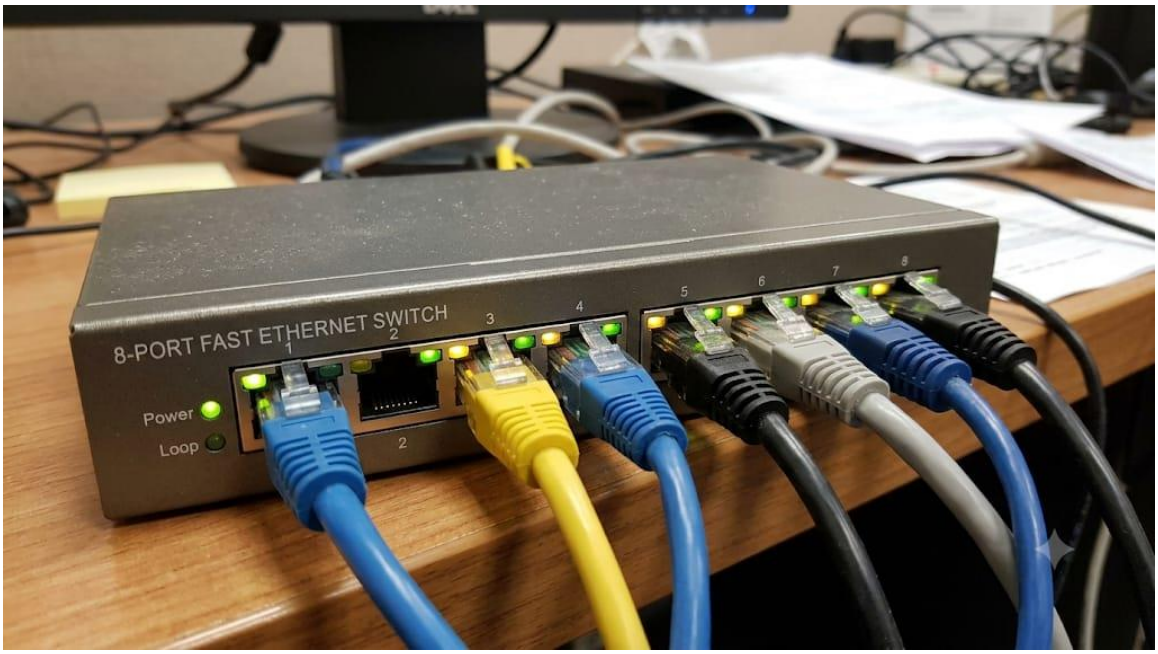


Identification and purpose:

- Power port (DC IN): Usually a round socket on the back. It connects the router to its power adapter.
- WAN / Internet port: Usually a separate, often blue, Ethernet port. It receives the Internet connection from the modem or ISP device.

- LAN 1, LAN 2 (and other LAN ports): Ethernet sockets used to connect a PC, printer or network switch.
- LED indicators: Usually on the front or top. Power shows that the router is on; Internet/WAN shows Internet connection status; Wi-Fi shows wireless status; LAN LEDs show wired connections and activity.

B. Network Switch — real product photo reference:



Identification and purpose:

- Power port: Usually on the back or side. It connects the switch to its power adapter.
- Ethernet ports 1 and 2: Two of the numbered RJ-45 sockets on the front. They connect PCs, routers, printers or other network devices. The other ports work in the same way.
- LED indicators: Small lights near each Ethernet port. A steady light usually means a link is connected; a blinking light usually means data is being transferred. A separate Power LED shows that the switch is on.

Question 4. Gaming café network scenario.

I am setting up a small gaming café with 8 gaming PCs. First, I will connect the Internet line from the ISP modem/ONT to the WAN port of the router. The router will provide Internet access to the café network and manage traffic between the local network and the Internet.

Next, I will connect one LAN port of the router to an Ethernet port on an 8-port (or larger) Gigabit network switch using a network cable. Then I will connect the 8 gaming PCs to the switch using separate Ethernet cables. This allows all PCs to communicate on the local network and access the Internet through the router.

I will use a Gigabit switch and good-quality Ethernet cables for stable, fast connections during IPL streaming and multiplayer gaming. The router is connected first to the ISP because it connects the café network to the Internet. The switch is used to connect many PCs together.

ISP Modem/ONT → Router (WAN) → Router (LAN) → Switch → 8 Gaming PCs

Conclusion

A router connects different networks and provides Internet access. A switch connects multiple devices within the same local network. Both devices are important for a reliable home or business network.